

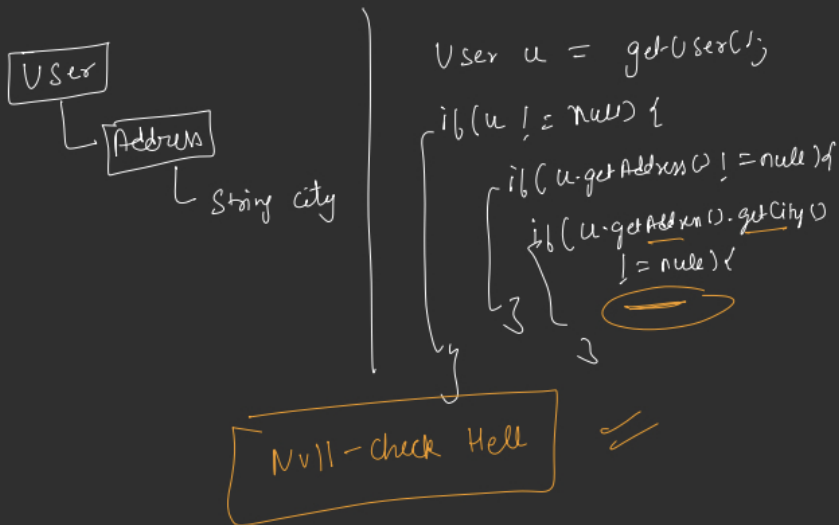
Optional classes

```
String name = getName();
```

```
if (name != null) {
```

```
    int len = name.length();
```

```
}
```

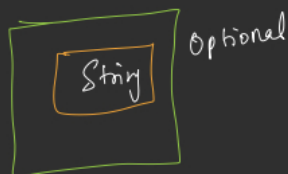
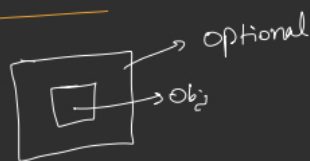


Problem with null

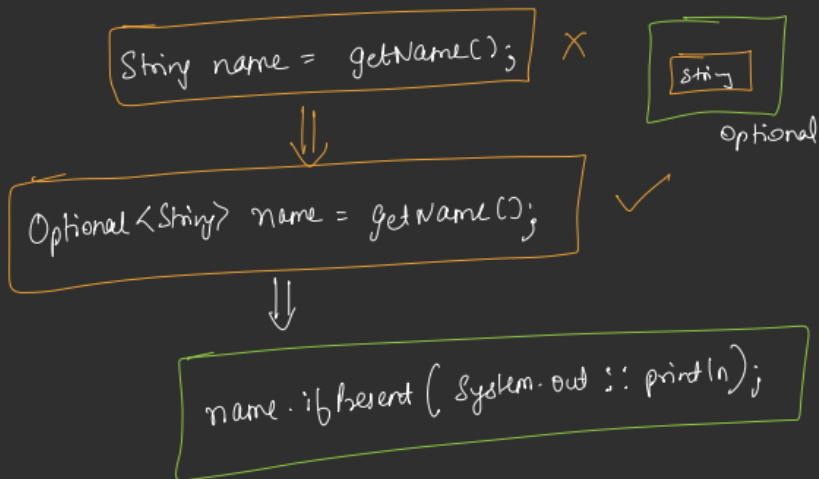
```
String name = getName();
```

- ↳ empty name
- ↳ error.
- ↳ expected.

Optional class :



Optional <T> → optional <String>



String name = null; (✗)

Optional<String> name = Optional.empty();

Optional<String> name = Optional.of("Aditya");

```
class Optional<T> {  
    private final T value;  
}
```

↳ (value == null) empty
↳ (value != null) present

Creating optional

Optional<String> n = Optional.of("Aditya");

Optional<String> n = Optional.ofNullable("Aditya");
null

⑦ = Optional.empty();

Getting value from optional

↳ get()

Optional<String> name = getName();

String name2 = name.get();

isPresent()

Optional<String> name = getName();

```
if (name.isPresent()) {  
    String s = name.get();  
}
```

ifPresent

name.ifPresent (System.out :: println);

→ orElse()

String = name.orElse ("Unknown");

→ orElseGet()

→ orElseThrow()

→ ifPresentOrElse()

Optional Transformation

↳ map()
↳ flatMap()
↳ filter()



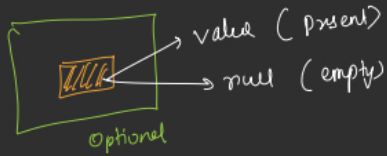
Optional<User> user = getUser();

```
user.map(x → x.getAddress())  
    .map(y → y.getCity())  
    .ifPresent(z → System.out.  
        println(z));
```

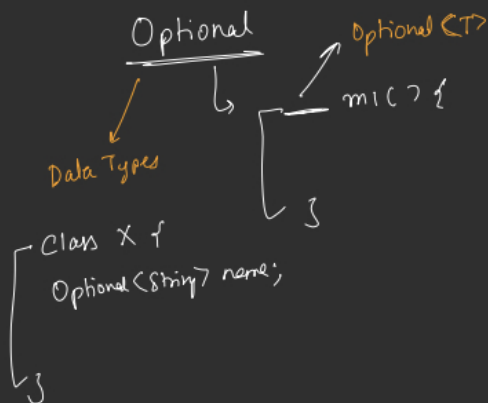
```
user.map (User :: getAddress)  
    .map (Address :: getCity)  
    .ifPresent (System.out :: println);
```

Optional = Mini Stream

↳ one value
↳ no value (null) → empty.



map()
filter()
flatMap()



Streams & optional

(0 to Many) (0 or 1 element)

Handle maybe value inside multiple values.

List <User> users;
↳ email (✓) } optional <String>
 (x)

```
users.stream()
  .map(u -> u.getEmail())
  .filter(x -> x.isPresent())
  .map(x -> x.get())
  .toList();
```

→ Stream <Optional <String>>